Exercise 01 - SQL Foundations

SQL fundamentals : SELECT and filtering

Selecting and Sorting

01
```
SELECT *
FROM employees;
```

id	first_name	last_name	department	salary	hire_date	city
1	Alice	Green	IT	70000	2020-01-10	Johannesburg
2	Brian	Lee	HR	450000	2019-03-22	Cape Town
3	Cathy	Zulu	Finance	65000	2018-07-18	Durban
4	David	Mokoena	Marketing	50000	2021-11-05	Pretoria
5	Eva	Naidoo	IT	72000	2017-09-30	Johannesburg

02
```
SELECT ~~Distinc~~ DISTINCT department
FROM employees;
```

department
IT
HR
Finance
Marketing

03
```
SELECT first_name, last_name
FROM employees
ORDER BY salary DESC;
```

first_name	last_name
Eva	Naidoo
Alice	Green
Cathy	Zulu
David	Mokoena
Brian	Lee

(04) SELECT id, first_name, last_name, salary
FROM employees
ORDER BY salary DESC
LIMIT 3;

id	first_name	last_name	salary
5	Eva	Naidoo	72000
1	Alice	Green	70000
3	Cathy	Zulu	65 000

Filtering with WHERE

(05) SELECT id, first_name, last_name, department
FROM employees
WHERE department = 'IT';

id	first_name	last_name	department
1	Alice	Green	IT
5	Eva	Naidoo	IT

(06) SELECT id, first_name, last_name, department, salary
FROM employees
WHERE salary > 60000 AND department = 'Finance';

id	first_name	last_name	department	salary
3	Cathy	Zulu	Finance	65000

(07) SELECT id, first_name, last_name, department
FROM employees
WHERE department = 'HR' OR 'Markerting';

id	first_name	last_name	department
2	Brian	Lee	HR
4	David	Mokoena	Marketing

08 SELECT id, first_name, last_name, department
FROM employees.
WHERE NOT department = 'IT';

id	first_name	last_name	department
2	Brian	Lee	HR
3	Cathy	Zulu	Finance
4	David	Mokoena	Marketing

09 SELECT id, first_name, last_name, department
FROM employees
WHERE IN department ('IT', 'HR', 'Finance');

id	first_name	last_name	department
1	Alice	Green	IT
2	Brian	Lee	HR
3	Cathy	Zulu	Finance
5	Eva	Naidoo	HR

10 SELECT id, first_name, last_name, department, salary, city
FROM employees
WHERE salary > 65000 AND city = 'Johannesburg', department
= 'IT';

id	first_name	last_name	department	salary	city
1	Alice	Green	IT	70000	Johannesburg
5	Eva	Naidoo	IT	72000	Johannesburg